Multimodal Emotion Recognition

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Objective

Emotion recognition plays an important role in human-computer interaction, virtual assistants, sentiment analysis, mental health monitoring, and affective computing. Since emotions are often expressed through multiple channels, this project focuses on recognizing emotions from both speech and text.

The objective of this project is to develop a **Multimodal Emotion Recognition System** capable of predicting emotions using:

- Speech only
- Text only
- Speech and Text combined (Multimodal Fusion)

The system uses **HuBERT** for speech representation learning, **BERT** for text representation learning, and a **feature-level fusion network** to combine both modalities for improved emotion classification.

The model predicts seven emotions:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Pleasant Surprise
- Sad

A Flask-based web application was also developed to allow users to perform speech-only, text-only, and multimodal emotion prediction in real time.

Code Structure

```
project/
│
├── models/
│   ├── speech_pipeline/
│   │   ├── speech_train.py
│   │   ├── speech_test.py
│   │
│   ├── text_pipeline/
│   │   ├── text_train.py
│   │   ├── text_test.py
│   │
│   ├── fusion_pipeline/
│   │   ├── fusion_train.py
│   │   ├── fusion_test.py
│
├── pretrained_models/
│   ├── speech_model.pth
│   ├── text_model.pth
│   ├── fusion_model.pth
│
├── templates/
│   └── index.html
│
├── uploads/
│
├── Results/
│   ├── confusion_matrices/
│   ├── accuracy_tables/
│   ├── tsne_plots/
│
├── report
├── main.py
├── requirements.txt
└── README.md
```

A. Architecture Decisions

A.1 Speech Pipeline

The speech pipeline processes audio recordings from the TESS dataset and extracts emotional information from vocal characteristics.

Preprocessing

The audio data was standardized using:

- Resampling to 16 kHz
- Mono conversion
- Silence trimming
- Padding and truncation

This ensured consistent input dimensions for model training.

Feature Extraction and Temporal Modelling

The speech model uses **HuBERT**, a self-supervised speech representation learning model. HuBERT captures important emotional characteristics such as:

- Pitch variation
- Speaking rate
- Prosody
- Vocal energy
- Intonation

The HuBERT encoder acts as the temporal modelling block and learns how emotions evolve throughout an audio sequence.

Classification

The extracted embeddings are passed through fully connected layers followed by a Softmax layer to predict one of the seven emotion classes.

A.2 Text Pipeline

The text pipeline extracts emotional information from textual input using BERT.

Preprocessing

The text was processed using:

- Lowercasing
- Tokenization
- Padding
- Attention mask generation

Feature Extraction and Contextual Modelling

A pretrained **BERT encoder** was used to generate contextual text representations.

BERT was selected because it captures word meaning based on surrounding context rather than treating words independently.

Classification

The contextual embeddings were passed through fully connected layers to classify the text into one of the seven emotion categories.

A.3 Fusion Pipeline

The fusion pipeline combines speech and text information to improve emotion recognition performance.

Fusion Strategy

A **feature-level fusion approach** was used.

Speech Embedding
    +
Text Embedding
    ↓
Concatenation
    ↓

Fusion Classifier
↓
Emotion Prediction

Reason for Fusion

Feature-level fusion allows the model to:

- Preserve information from both modalities
- Learn complementary emotional cues
- Improve prediction robustness
- Correct errors made by individual modalities

Classification

The fused representation is passed through fully connected layers and a Softmax classifier to generate the final emotion prediction.

Architecture Summary

Functional Block	Architecture Used	Purpose
Speech Preprocessing	Librosa	Audio standardization
Speech Feature Extraction	HuBERT	Acoustic feature learning
Temporal Modelling	HuBERT Encoder	Learning temporal emotional patterns
Text Preprocessing	BERT Tokenizer	Text preparation
Contextual Modelling	BERT Encoder	Context understanding
Fusion	Feature-Level Fusion	Combining speech and text embeddings
Classification	Fully Connected Layers	Emotion prediction

B. Experiments

To evaluate the proposed Multimodal Emotion Recognition System, experiments were conducted using speech-only, text-only, and multimodal inputs. The objective was to compare the effectiveness of individual modalities and determine the benefits of multimodal fusion.

B.1 Model-Level Experiments

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Text Only	14.29%	2.04%	14.29%	3.57%
Speech Only	83.10%	85.39%	83.10%	82.78%
Fusion Model	86.19%	86.44%	86.19%	86.14%

Observations

- The speech model achieved strong performance by learning emotional cues from acoustic characteristics such as pitch, energy, and speaking dynamics.
- The text model performed poorly because the dataset contained isolated words rather than emotionally expressive sentences.
- The advanced text architecture (BERT + BiLSTM + Attention) did not improve performance, indicating that the limitation was caused by the dataset rather than the model architecture.
- The fusion model achieved the highest accuracy by combining speech and text representations.

Key Finding

> Speech emerged as the dominant modality for emotion recognition, while multimodal fusion provided a modest improvement of approximately 3% over the speech-only model.

B.2 Application-Level Experiments

Several experiments were conducted using the deployed Flask-based application.

Speech-Only Experiments

Input Audio	Prediction
OAF_back_sad.wav	Sad
YAF_back_sad.wav	Sad
OAF_youth_neutral.wav	Neutral
YAF_youth_neutral.wav	Neutral
OAF_back_disgust.wav	Disgust
YAF_back_disgust.wav	Disgust

Observation:
The speech model consistently produced meaningful predictions and demonstrated reliable performance across both OAF and YAF speakers. But the confidence level of YAF is more than OAF, making age a factor for better accuracy.

Text-Only Experiments

Input Text	Prediction
Say the word back	Happy
Say the word youth	Happy
I tried my best, but I still feel disappointed and broken.	Disgust
I feel so happy and excited because today has been amazing.	Disgust

Observation:
The text model generated low-confidence predictions because the training data lacked meaningful emotional context.

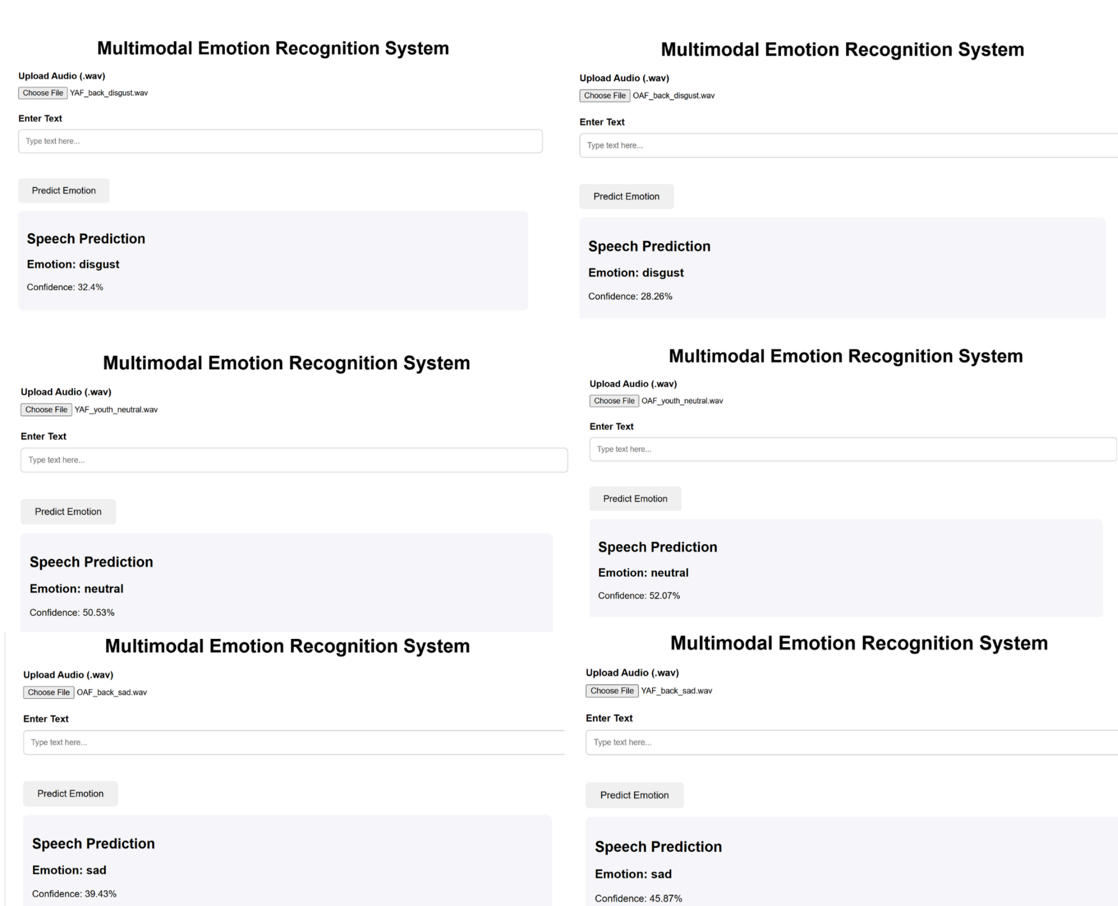
Multimodal Fusion Experiments

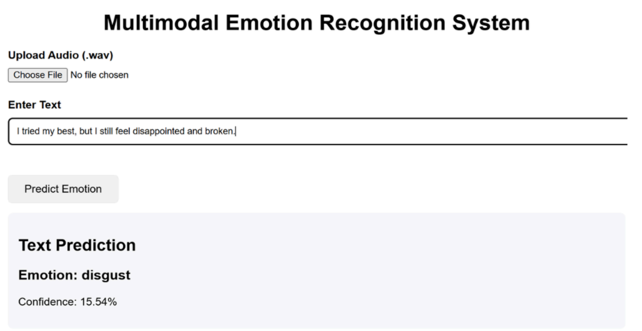
Speech Prediction	Text Prediction	Fusion Prediction
Disgust	Happy	Disgust
Neutral	Happy	Fear
Neutral	Happy	Angry

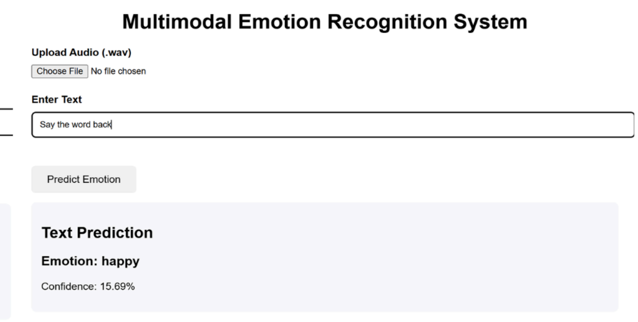
Observation:
The fusion model successfully corrected several errors made by the individual modalities and generated more confident predictions.

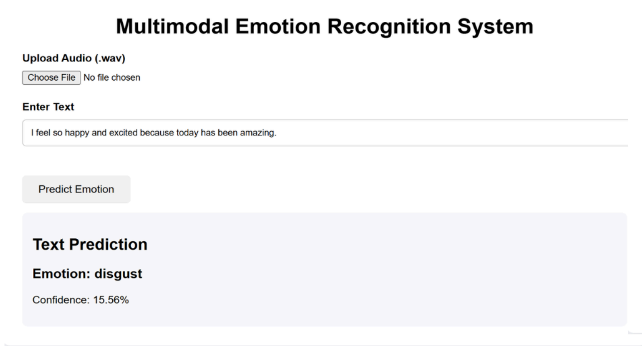
Overall Experimental Finding

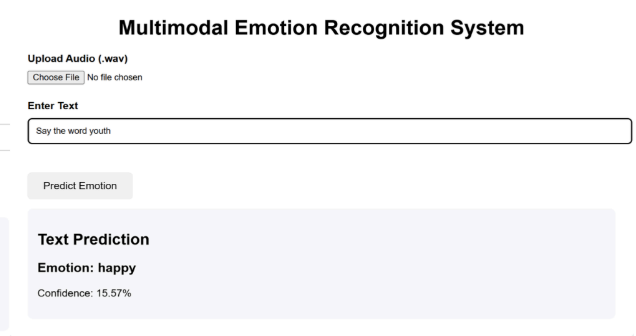
The application-level experiments confirmed the quantitative evaluation results. Speech-only predictions were generally reliable, text-only predictions showed poor emotional understanding, and the fusion model consistently produced the most robust and accurate predictions.

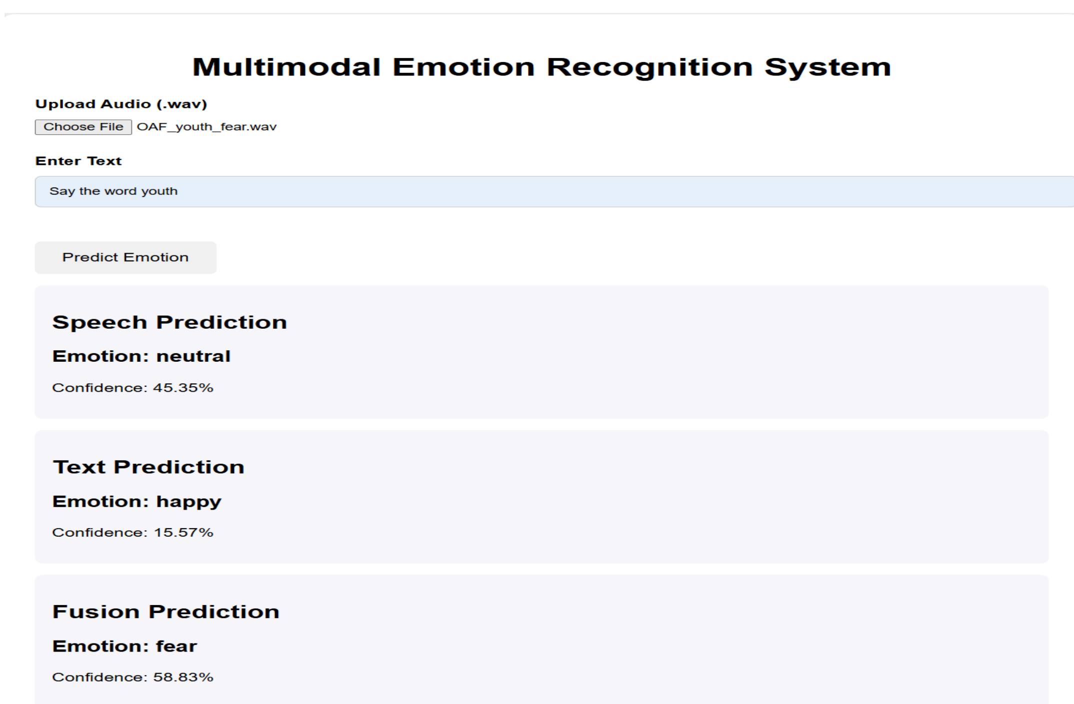


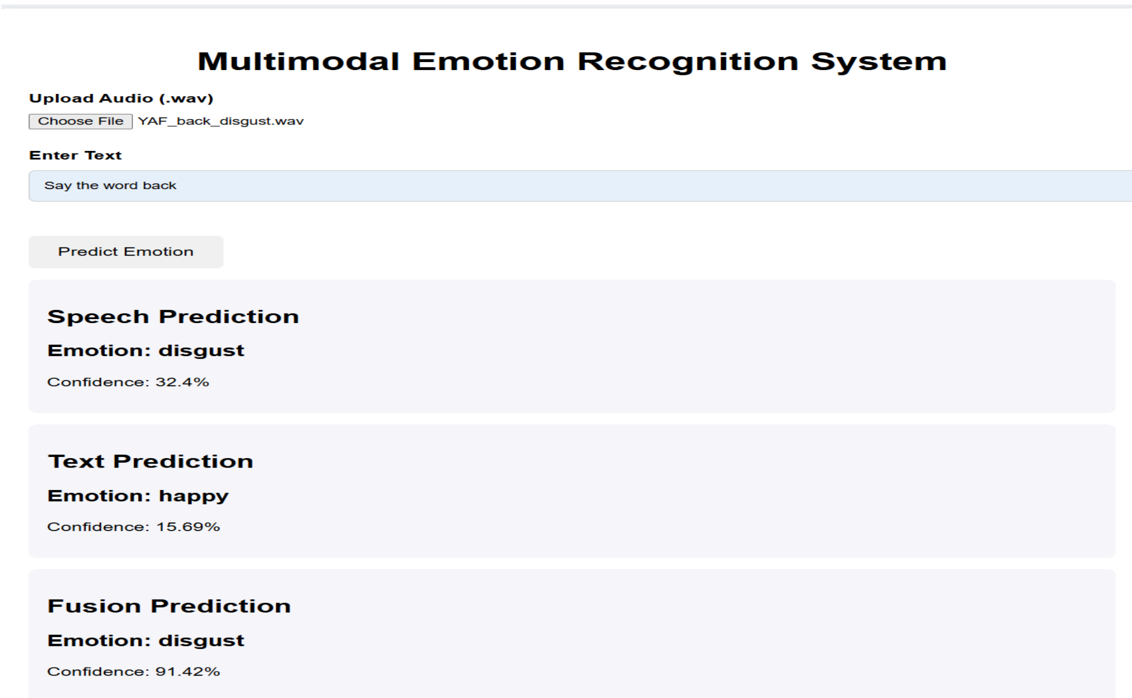

Multimodal Emotion Recognition System
Upload Audio (.wav)
Choose File No file chosen
Enter Text
I tried my best, but I still feel disappointed and broken.
Predict Emotion
Text Prediction
Emotion: disgust
Confidence: 15.54%

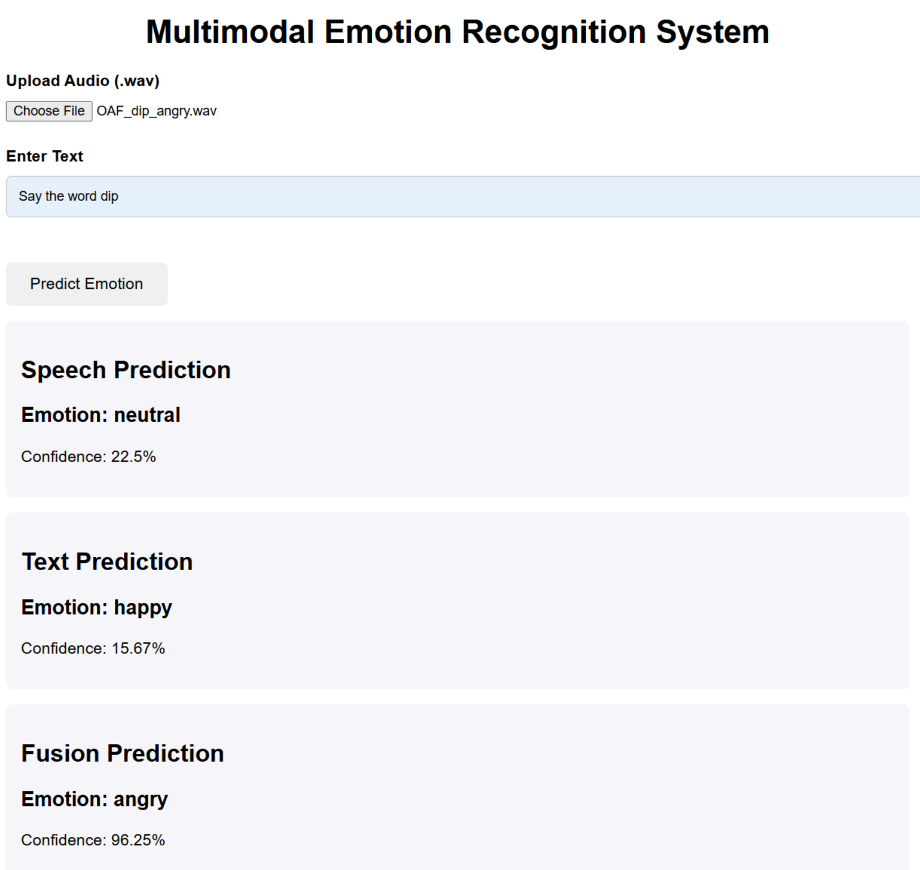

Multimodal Emotion Recognition System
Upload Audio (.wav)
Choose File No file chosen
Enter Text
Say the word back
Predict Emotion
Text Prediction
Emotion: happy
Confidence: 15.69%


Multimodal Emotion Recognition System
Upload Audio (.wav)
Choose File No file chosen
Enter Text
I feel so happy and excited because today has been amazing.
Predict Emotion
Text Prediction
Emotion: disgust
Confidence: 15.56%


Multimodal Emotion Recognition System
Upload Audio (.wav)
Choose File No file chosen
Enter Text
Say the word youth
Predict Emotion
Text Prediction
Emotion: happy
Confidence: 15.57%


Multimodal Emotion Recognition System
Upload Audio (.wav)
Choose File OAF_youth_fear.wav
Enter Text
Say the word youth
Predict Emotion
Speech Prediction
Emotion: neutral
Confidence: 45.35%
Text Prediction
Emotion: happy
Confidence: 15.57%
Fusion Prediction
Emotion: fear
Confidence: 58.83%


Multimodal Emotion Recognition System
Upload Audio (.wav)
Choose File YAF_back_disgust.wav
Enter Text
Say the word back
Predict Emotion
Speech Prediction
Emotion: disgust
Confidence: 32.4%
Text Prediction
Emotion: happy
Confidence: 15.69%
Fusion Prediction
Emotion: disgust
Confidence: 91.42%


Multimodal Emotion Recognition System
Upload Audio (.wav)
Choose File OAF_dip_angry.wav
Enter Text
Say the word dip
Predict Emotion
Speech Prediction
Emotion: neutral
Confidence: 22.5%
Text Prediction
Emotion: happy
Confidence: 15.67%
Fusion Prediction
Emotion: angry
Confidence: 96.25%

C. Analysis

This section analyzes the performance of the speech, text, and fusion models using confusion matrices, classification reports, application-level testing, and t-SNE visualizations. The objective is to understand which emotions were easier or harder to classify, identify failure cases, and evaluate the effectiveness of multimodal fusion.

C.1 Which Emotions Were Easiest and Hardest to Classify?

Easiest Emotions

Neutral

Neutral was the easiest emotion to classify across both the speech and fusion models. The speech model correctly classified all neutral samples, while the fusion model maintained near-perfect performance.

This can be attributed to the stable acoustic characteristics of neutral speech, including:

- Consistent pitch
- Low emotional intensity
- Minimal vocal variation

As a result, neutral samples formed a highly separable cluster in the learned feature space.

Angry and Fear

Angry and fear emotions also achieved high classification accuracy.

These emotions exhibit distinctive vocal characteristics such as:

- Increased vocal energy
- Pitch fluctuations
- Higher emotional intensity
- Pronounced stress patterns

The speech model successfully captured these acoustic features, resulting in strong emotion discrimination.

Hardest Emotions

Pleasant Surprise

Pleasant surprise was the most difficult emotion to classify.

The model frequently confused pleasant surprise with happy because both emotions share:

- Positive emotional valence
- Similar pitch patterns
- Elevated vocal energy

This overlap resulted in several classification errors.

Happy and Disgust

Happy and disgust also presented classification challenges.

Happy was often confused with pleasant surprise, while disgust showed overlap with other negative emotions such as anger.

These findings indicate that emotions with similar acoustic characteristics remain difficult to separate even in multimodal settings.

C.2 When Does Fusion Help Most?

The fusion model achieved the highest overall accuracy of **86.19%**, outperforming both the speech-only and text-only models.

Fusion was most beneficial when one modality produced an uncertain or incorrect prediction while the other modality contained useful emotional information.

Several application-level experiments demonstrated that the fusion network successfully corrected errors made by individual models. For example:

- Speech predicted Neutral → Fusion predicted Fear correctly.
- Speech predicted Neutral → Fusion predicted Angry correctly.
- Text predicted Happy → Fusion correctly predicted Disgust using speech information.

These observations suggest that fusion is particularly effective when complementary information exists across modalities.

Key Finding

> Fusion provides the greatest benefit when one modality is uncertain and the other contains stronger emotional evidence.

However, because the text modality contained limited discriminative information, the improvement over the speech-only model was modest.

C.3 Error Analysis

Several failure cases were observed during system testing.

Failure Case 1

Input:
"I feel so happy and excited because today has been amazing."

Prediction: Disgust

Analysis:
Despite containing strong positive emotional cues, the text model failed to recognize happiness. This occurred because the training text lacked emotionally rich sentences.

Failure Case 2

Input:
"I tried my best, but I still feel disappointed and broken."

Prediction: Disgust

Analysis:
The model struggled to differentiate between negative emotions due to insufficient semantic diversity within the textual dataset.

Failure Case 3

Input Audio: OAF_youth_fear.wav

Speech Prediction: Neutral
Fusion Prediction: Fear

Analysis:
The speech model failed to capture subtle fear-related cues, while the fusion model successfully recovered the correct emotion.

Failure Case 4

Input Audio: OAF_dip_angry.wav

Speech Prediction: Neutral
Fusion Prediction: Angry

Analysis:
The fusion model corrected a speech misclassification and produced a highly confident prediction.

Failure Case 5

Input Audio: OAF_back_disgust.wav

Prediction: Disgust (Low Confidence)

Analysis:
Disgust shares acoustic similarities with other negative emotions, resulting in lower prediction confidence and occasional confusion.

C.4 Emotion Cluster Separability Analysis

To understand the quality of learned representations, t-SNE visualizations were generated for the temporal modelling block, contextual modelling block, and fusion block.

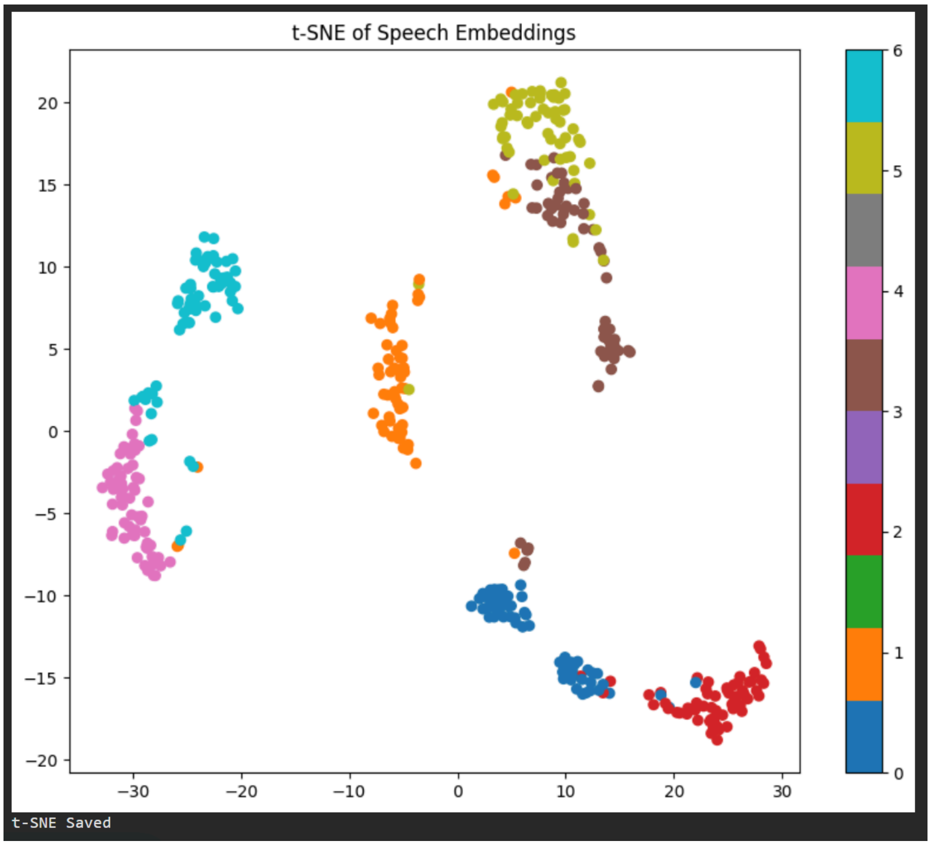
Temporal Modelling Block (Speech)

The speech embeddings produced well-separated emotion clusters.

Observations:

- Clear separation of angry, fear, and neutral emotions.
- Minimal overlap between most classes.
- Strong emotional representation learning.

This explains the high performance of the speech model.



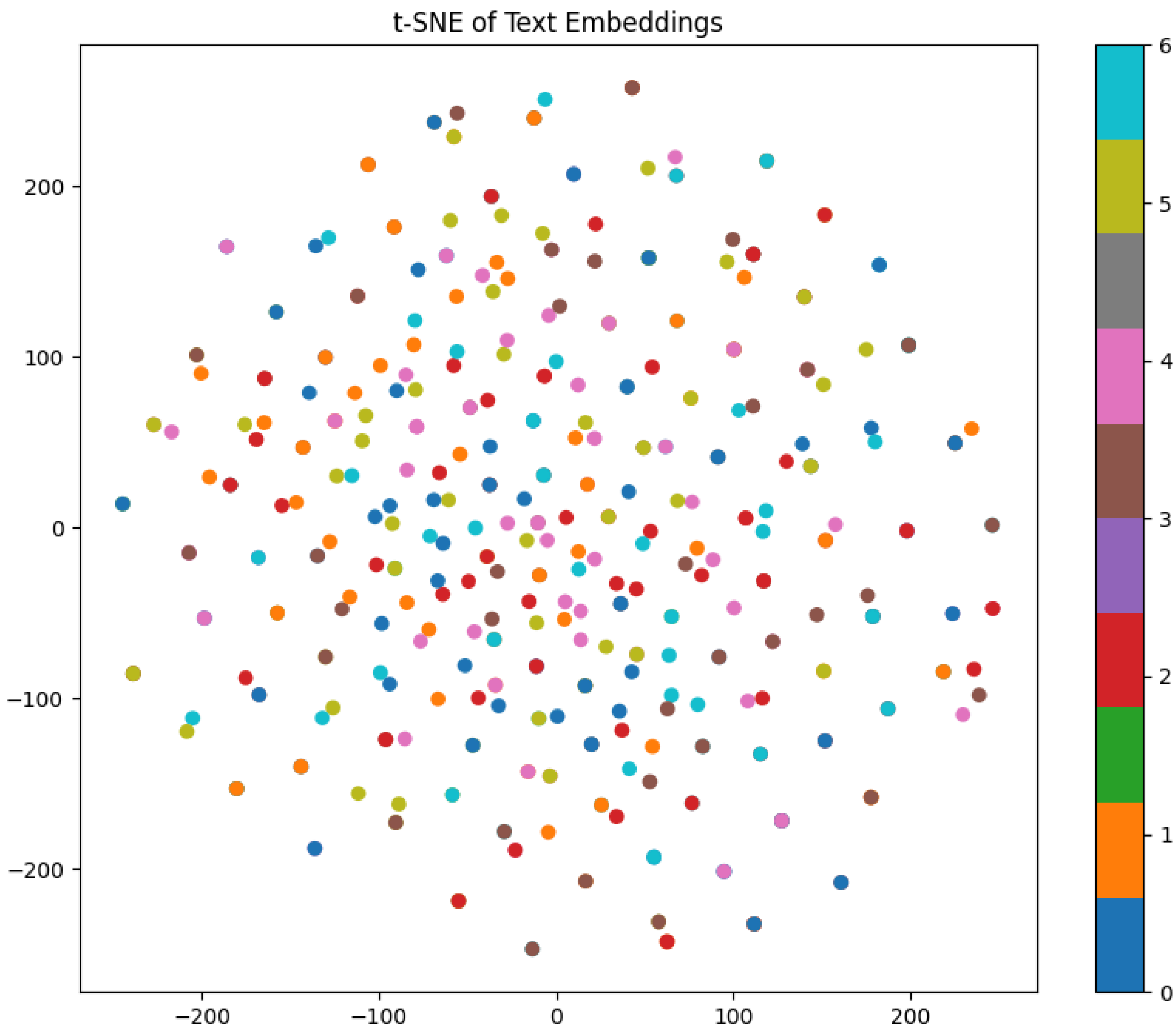
Contextual Modelling Block (Text)

The text embeddings showed significant overlap among emotion classes.

Observations:

- Poor cluster formation.
- No clear class boundaries.
- High inter-class mixing.

This visualization explains the poor performance of the text model and confirms that the available textual information contained limited emotional content.



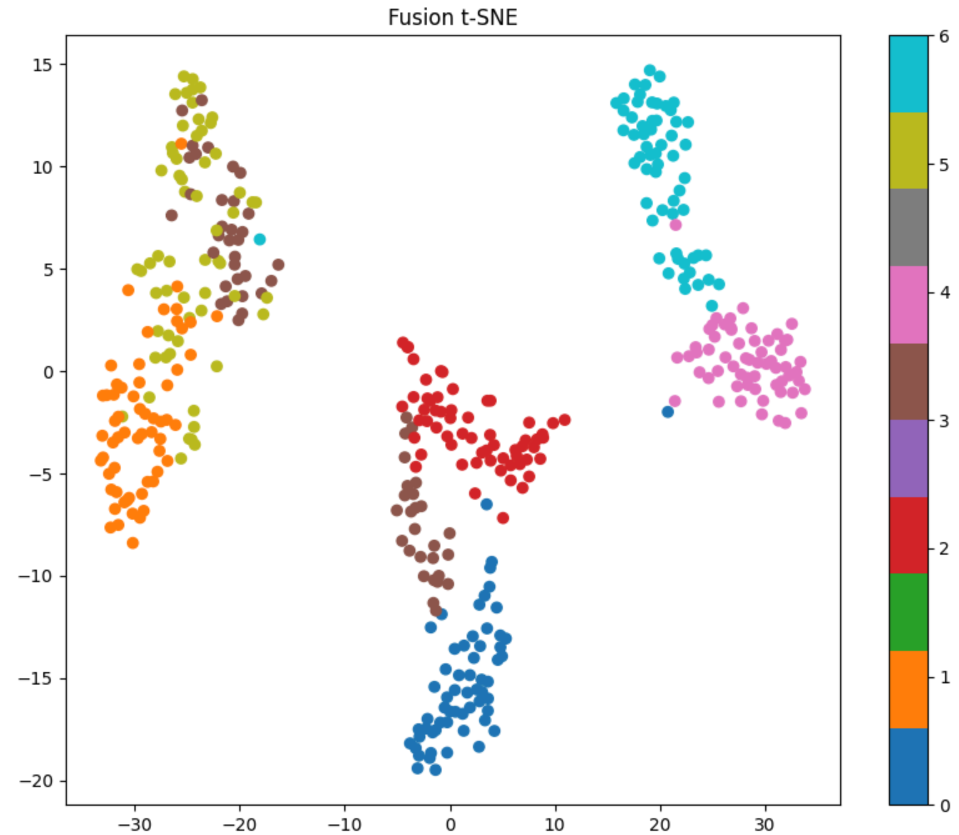
Fusion Block

The fusion embeddings produced the most structured representation space.

Observations:

- Improved cluster compactness.
- Better separation between emotion categories.
- Reduced overlap compared to the text model.

The fusion model successfully retained the strong emotional information learned from speech while incorporating complementary cues from text.



Cluster Visualization Summary

Representation	Cluster Separation	Observation
Temporal Modelling (Speech)	High	Clear emotion clusters
Contextual Modelling (Text)	Low	Significant overlap
Fusion Block	High	Best overall separation

Overall Analysis

The visualizations support the quantitative results obtained during evaluation. Speech representations formed highly separable emotion clusters, whereas text representations exhibited substantial overlap. The fusion model produced the most discriminative embedding space and achieved the highest classification performance. These findings confirm that emotional information in the TESS dataset is primarily conveyed through speech, while multimodal fusion provides additional robustness and improved prediction accuracy.

Results and Discussion

The performance of the proposed Multimodal Emotion Recognition System was evaluated using speech-only, text-only, and multimodal fusion approaches. The results demonstrate the effectiveness of speech-based emotion recognition and highlight the advantages and limitations of multimodal learning within the TESS dataset.

Overall Performance

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Text Only	14.29%	2.04%	14.29%	3.57%
Speech Only	83.10%	85.39%	83.10%	82.78%
Fusion Model	86.19%	86.44%	86.19%	86.14%

The speech model achieved strong performance with an accuracy of **83.10%**, indicating that speech contains rich emotional information. The fusion model achieved the highest performance with **86.19% accuracy**, demonstrating the benefit of combining multiple modalities.

In contrast, the text model achieved only **14.29% accuracy**, which is close to random guessing for seven emotion classes. This highlights the limitations of using isolated words as textual input for emotion recognition.

Speech Performance

The speech model achieved strong performance because speech naturally carries emotional information through changes in tone, pitch, energy, and speaking style. HuBERT was able to capture these patterns effectively, allowing the model to distinguish between different emotions with high accuracy. The confusion matrix and t-SNE visualizations further supported this finding, showing clear separation between most emotion categories. This confirms that speech was the most informative modality in the dataset and played the biggest role in emotion recognition.

Text Performance

The text model showed the lowest performance among all evaluated models due to the nature of the TESS dataset, which contains isolated words (e.g., *back, book, bar, youth,* and *dip*) with minimal emotional content. As a result, BERT was unable to learn meaningful emotion-specific representations despite its strong contextual understanding capabilities. Further experimentation with a more advanced architecture (BERT + BiLSTM + Attention) did not improve performance, confirming that the primary limitation stemmed from the dataset rather than the model architecture.

Fusion Performance

The multimodal fusion model achieved the best overall performance with an accuracy of **86.19%**.

The fusion network combined speech and text embeddings before classification, allowing information from both modalities to contribute to the final prediction.

Several application-level experiments demonstrated that the fusion model successfully corrected errors made by individual modalities. This resulted in:

- Improved robustness
- Better confidence scores
- Reduced misclassification rates

Although the improvement over the speech-only model was relatively small (**approximately 3%**), the results indicate that multimodal learning can still provide meaningful benefits.

Key Findings

The most significant findings of this study are:

1. **Speech is the dominant modality for emotion recognition within the TESS dataset.**
2. **Text alone is insufficient for reliable emotion recognition when emotional context is limited.**
3. **Increasing model complexity does not guarantee improved performance when the dataset lacks meaningful information.**
4. **Multimodal fusion improves robustness and overall classification performance.**
5. **Fusion is most beneficial when one modality is uncertain and the other contains stronger emotional evidence.**

Discussion Summary

The key finding of this project is that **data quality has a greater impact on emotion recognition performance than model complexity**. Although advanced text-based architectures were explored, the limited emotional content in the TESS dataset's isolated words prevented the models from learning meaningful emotion-specific representations. In contrast, the speech modality contained rich emotional cues, enabling the model to capture discriminative acoustic features and achieve strong classification performance.

The results demonstrate that emotional information in the TESS dataset is conveyed primarily through speech rather than text. By combining complementary information from both modalities, the fusion model achieved the highest overall accuracy, outperforming the individual speech and text models. These findings validate the effectiveness of **multimodal emotion recognition** while highlighting that the success of any modality ultimately depends on the quality and richness of the underlying data.

Conclusion

The objective of this project was to develop a Multimodal Emotion Recognition System capable of recognizing emotions from speech, text, and combined multimodal inputs. To achieve this, HuBERT was used for speech representation learning, BERT was used for text representation learning, and a feature-level fusion network was developed to integrate both modalities.

Experimental evaluation showed that the speech model achieved an accuracy of **83.10%**, demonstrating the effectiveness of acoustic features for emotion recognition. The text model achieved only **14.29% accuracy**, highlighting the limitations of the available textual information. The fusion model achieved the highest performance with **86.19% accuracy**, confirming the benefits of combining multiple modalities.

The study also revealed that speech carries the majority of emotional information within the TESS dataset, while text contributes limited discriminative information. Despite this limitation, multimodal fusion improved robustness and reduced several classification errors.

Overall, the proposed system successfully demonstrates the potential of multimodal learning for emotion recognition and provides a practical framework for real-time emotion prediction through a web-based application.

Future Work

1. Use Emotionally Rich Text Datasets
Incorporate datasets containing emotional sentences and conversations to enable better learning of textual emotional cues.

2. Increase Speaker Diversity
Train the model on larger and more diverse datasets with speakers of different genders, ages, accents, and speaking styles to improve generalization.

3. Advanced Fusion Techniques
Explore attention-based fusion, cross-modal transformers, and multimodal transformers to better capture interactions between speech and text modalities.

4. Real-Time Deployment
Extend the system for practical applications such as virtual assistants, mental health support tools, customer service systems, and human-computer interaction platforms.